Prediction Market Analysis

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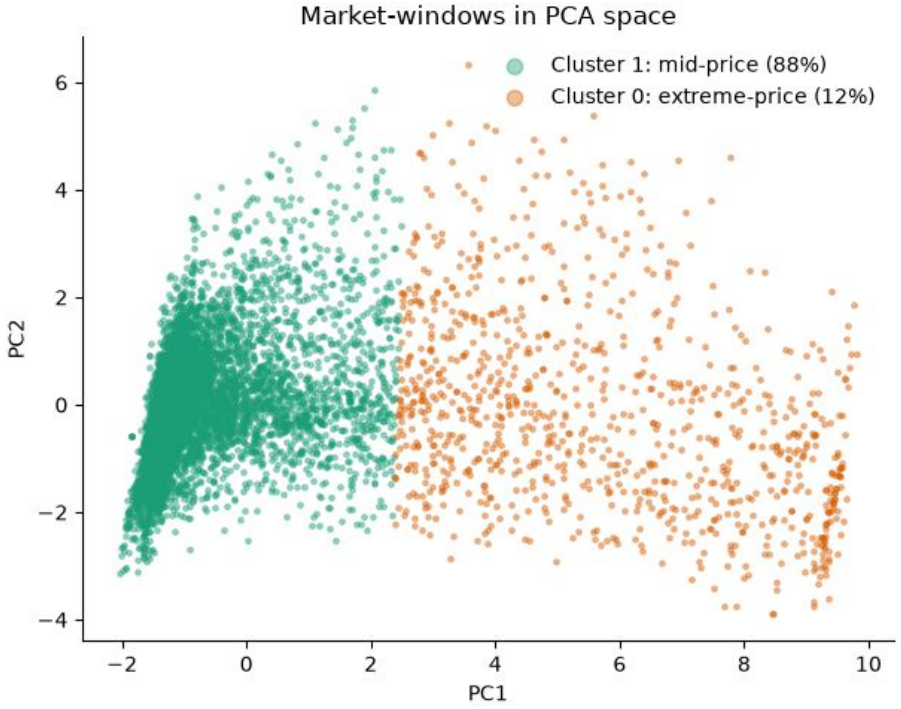
Github Repository

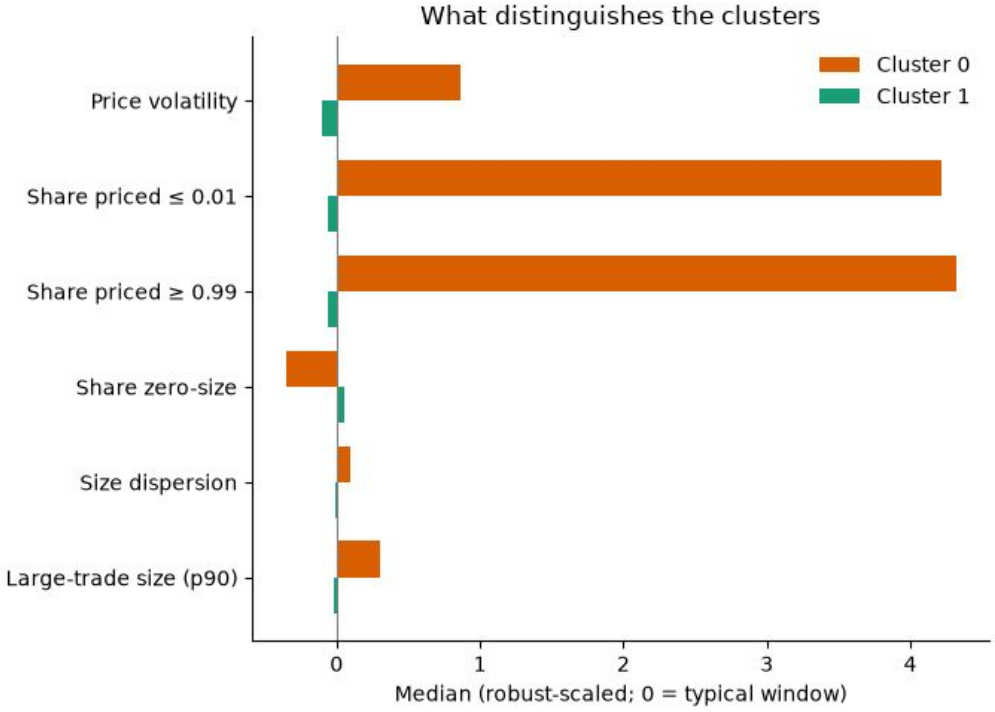
Polymarket

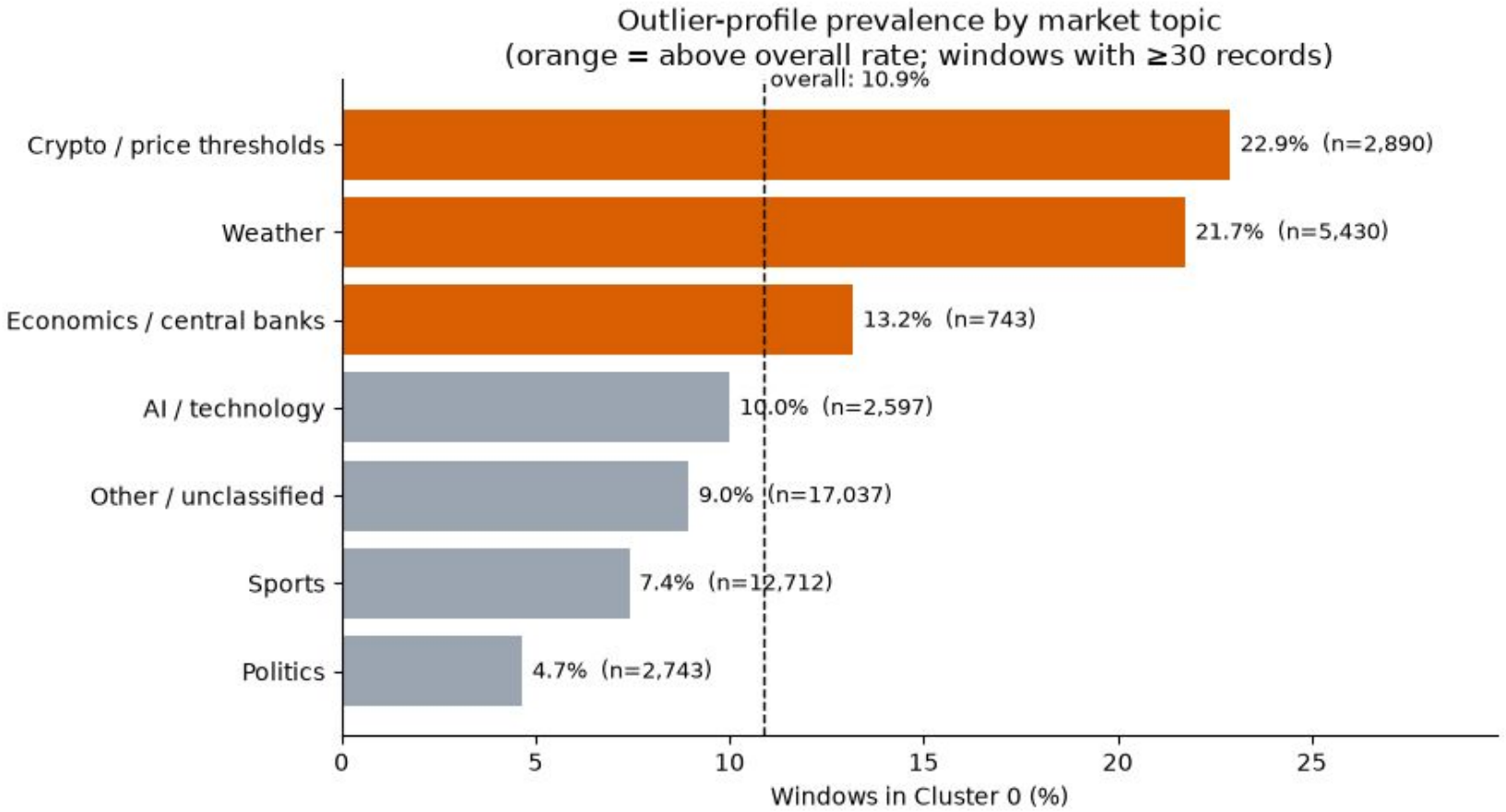
Investigating Behavioral Patterns

- Looking at whether clustering can reveal distinct market behaviors
- Process
  - Unit of analysis: Market Window
    - Captures localized changes
  - Engineered Features
    - Price sd, near 0/1 fraction, zero size fraction, size distribution
  - K-means: $k = 2$
    - Strong separation under sample

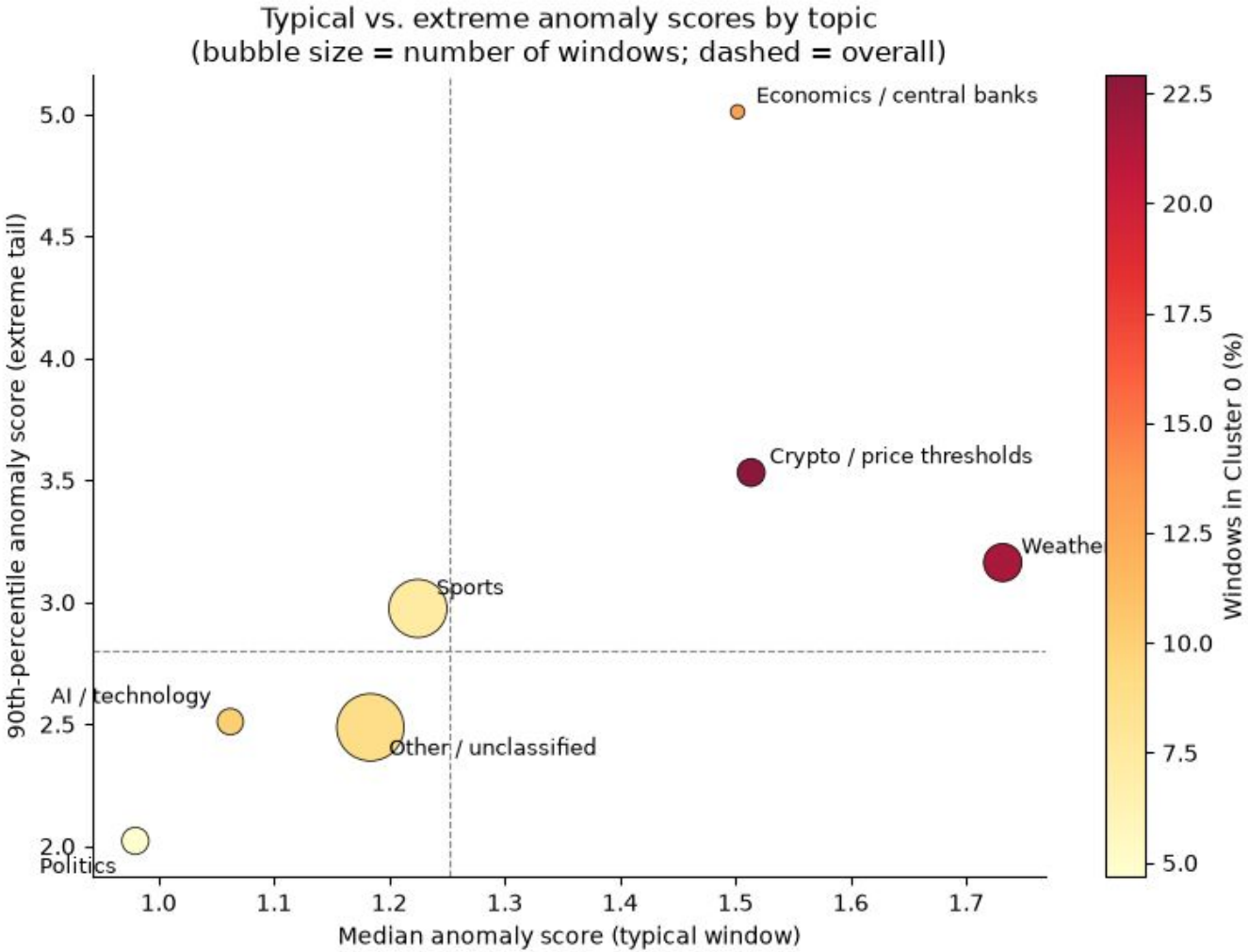
Two Clusters







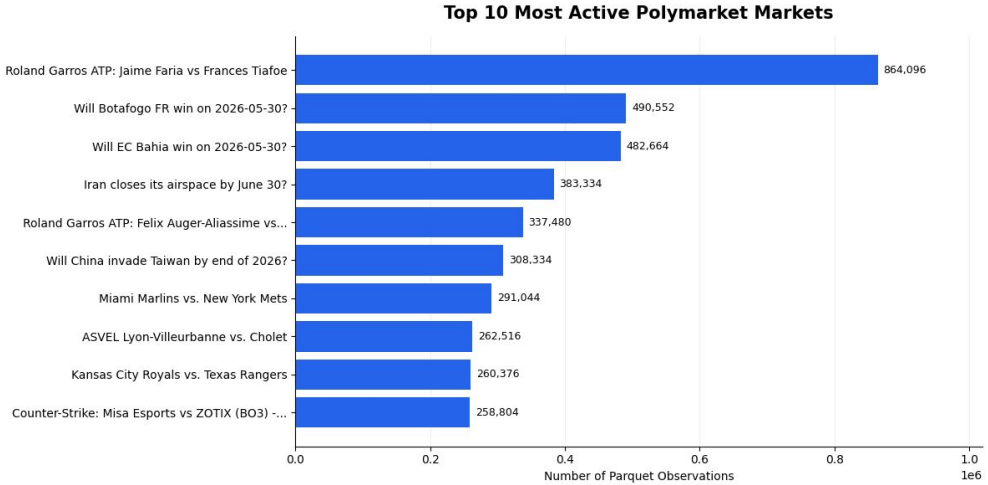
Market categories derived from market titles, keyword-based title classification


Typical vs. extreme anomaly scores by topic
(bubble size = number of windows; dashed = overall)
Economics / central banks
Crypto / price thresholds
Weathe
Sports
AI / technology
Other / unclassified
Politics
90th-percentile anomaly score (extreme tail)
Median anomaly score (typical window)
Windows in Cluster 0 (%)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.0
1.1
1.2
1.3
1.4
1.5
1.6
1.7
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0

Exploring Parquet Order Book Data

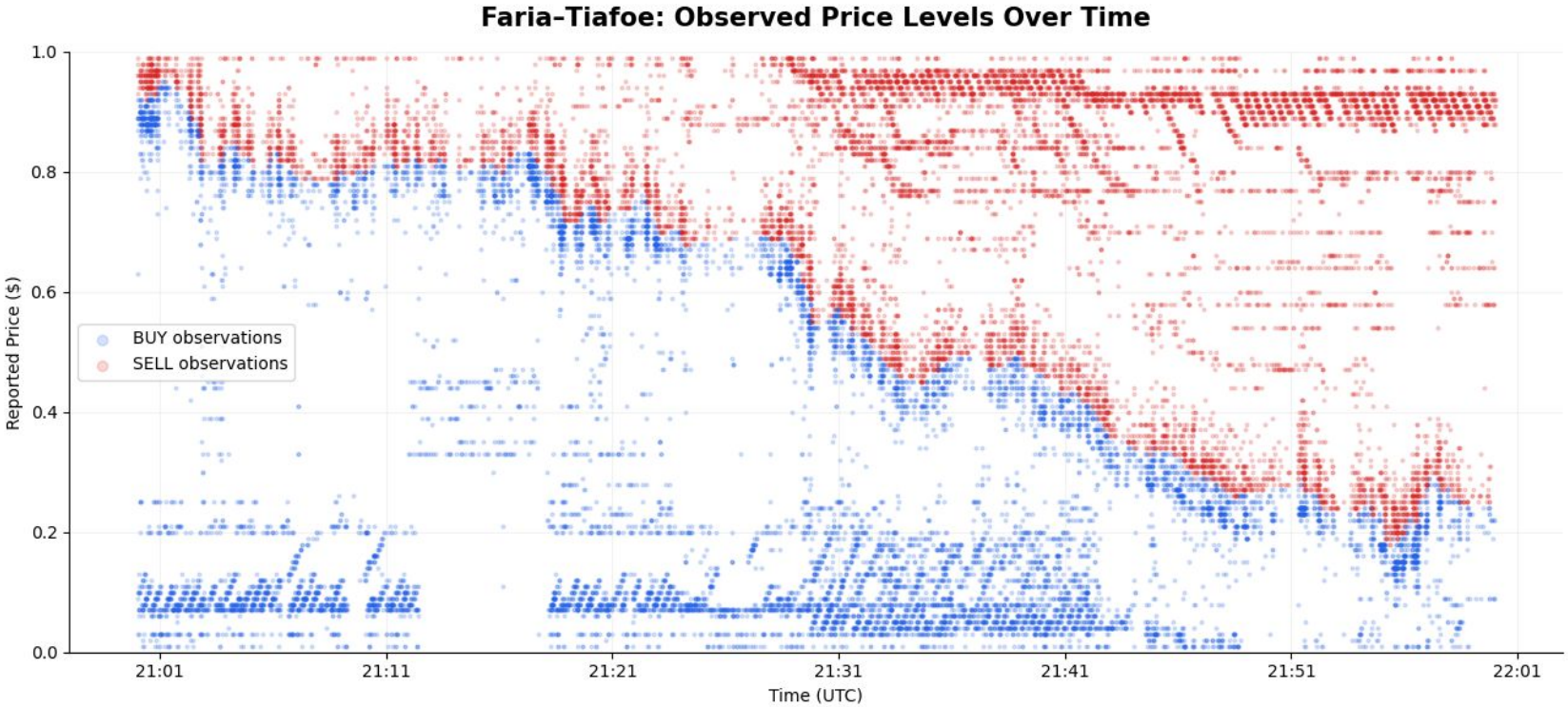
One-hour sample | May 30th, 2026 | 21:00 - 22:00 UTC

- Records capture timestamped BUY and SELL price levels and associated quantities, providing insight into market activity and liquidity
- Faria vs. Tiafoe was selected for deeper analysis because it had the highest observation count, allowing us to investigate how its order book evolved



Key Challenge: Individual observations do not necessarily represent trades or complete order book snapshots

Observed Price Movements



Observed price levels ≠ reconstructed best bid/ask

What does a parquet row actually represent?

Initial assumption: Each row represents an update that can be replayed to reconstruct the historical order book

Difficulties

- Ambiguous observations (trades, individual orders, incremental updates)
- Event sequencing —>
- Approximately 17% of observations have size 0 maybe indicating removed price labels

Timestamp (UTC)	Side	Price	Size
21:00:00.39	BUY	$0.91	4,951.40
21:00:00.39	BUY	$0.91	4,768.40
21:00:00.39	BUY	$0.91	4,368.40
21:00:00.39	BUY	$0.91	4,318.40

Attempting Order Book Reconstruction

Our approach:

- Filter Faria-Tiafoe observations to a single asset id (432,048 rows)
- Use the physical parquet row order to break timestamp ties
- Maintain the separate BUY and SELL dictionaries
- Replace quantities for positive sizes
- Calculate the spread using the highest retained bid and lowest retained ask

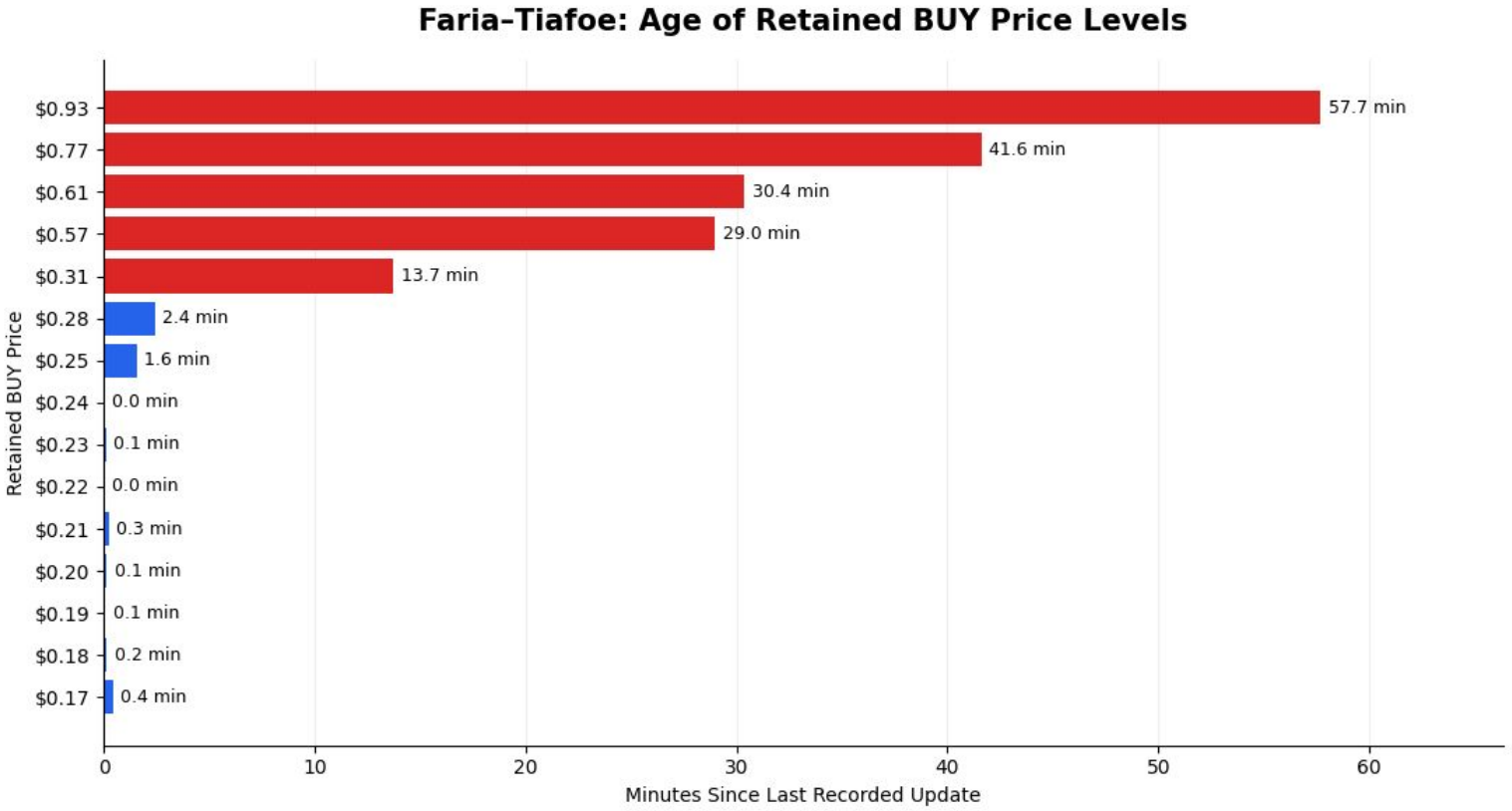
Important Assumptions:

- Reported sizes represent the total available quantity ata price level, rather than a change in quantity
- All relevant changes to the order book are captured in the dataset
- Physical row order reflects actual sequence of updates
- Start with empty order book

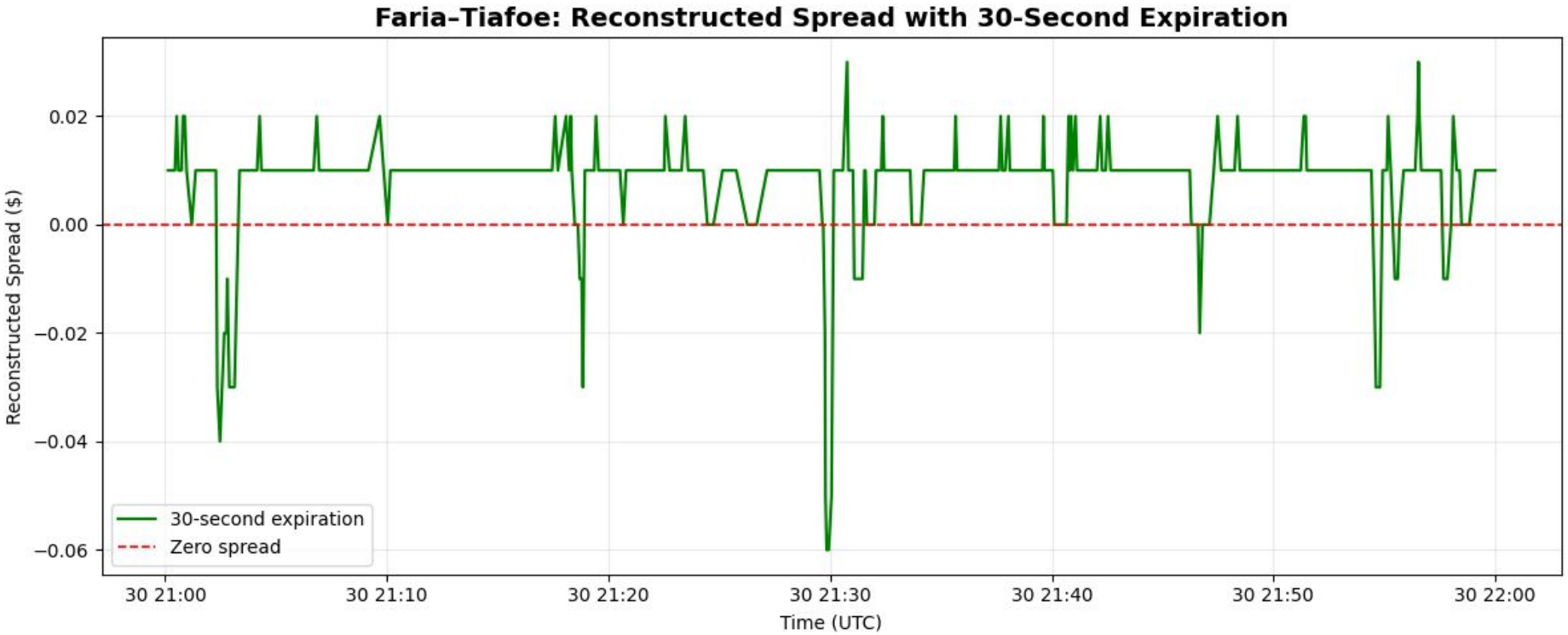
Negative Bid-Ask Spreads



Examining Retained Price Levels



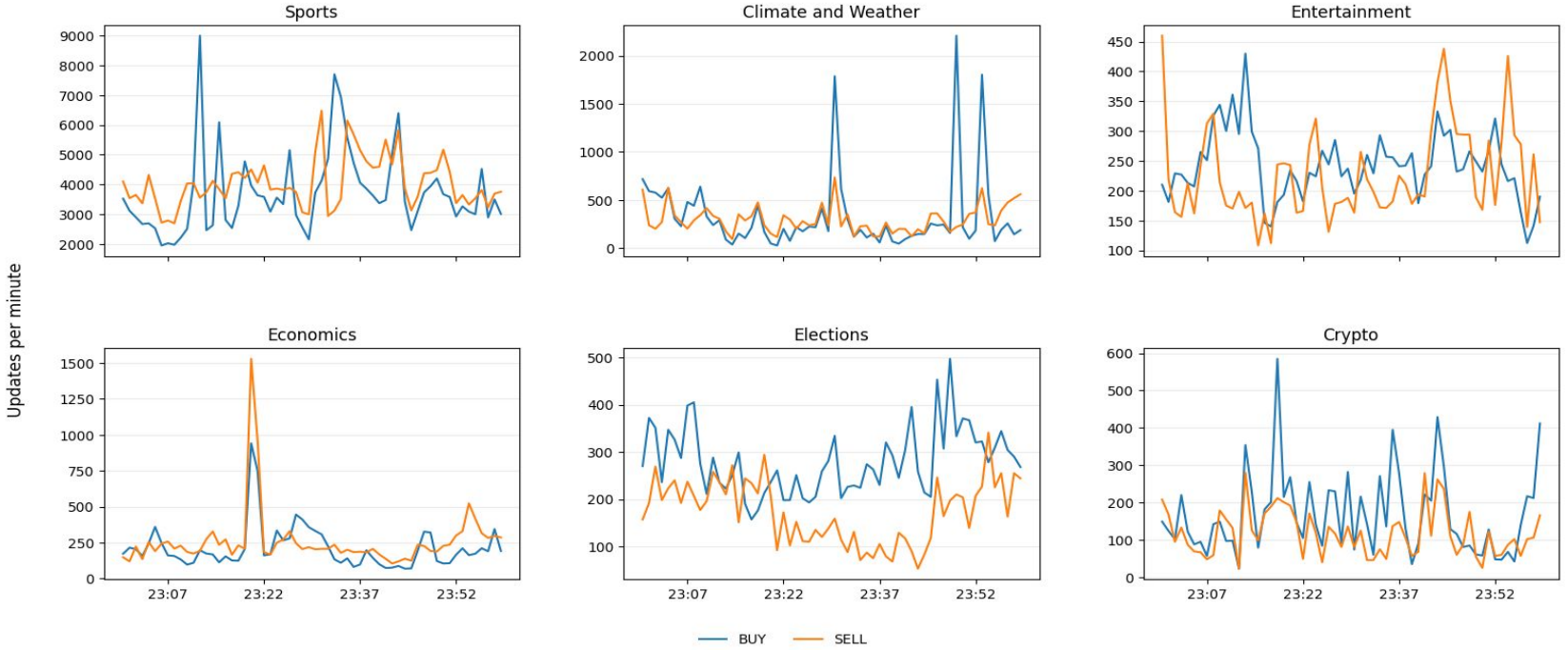
Improving Reconstruction with Price Level Expiration



Kalshi

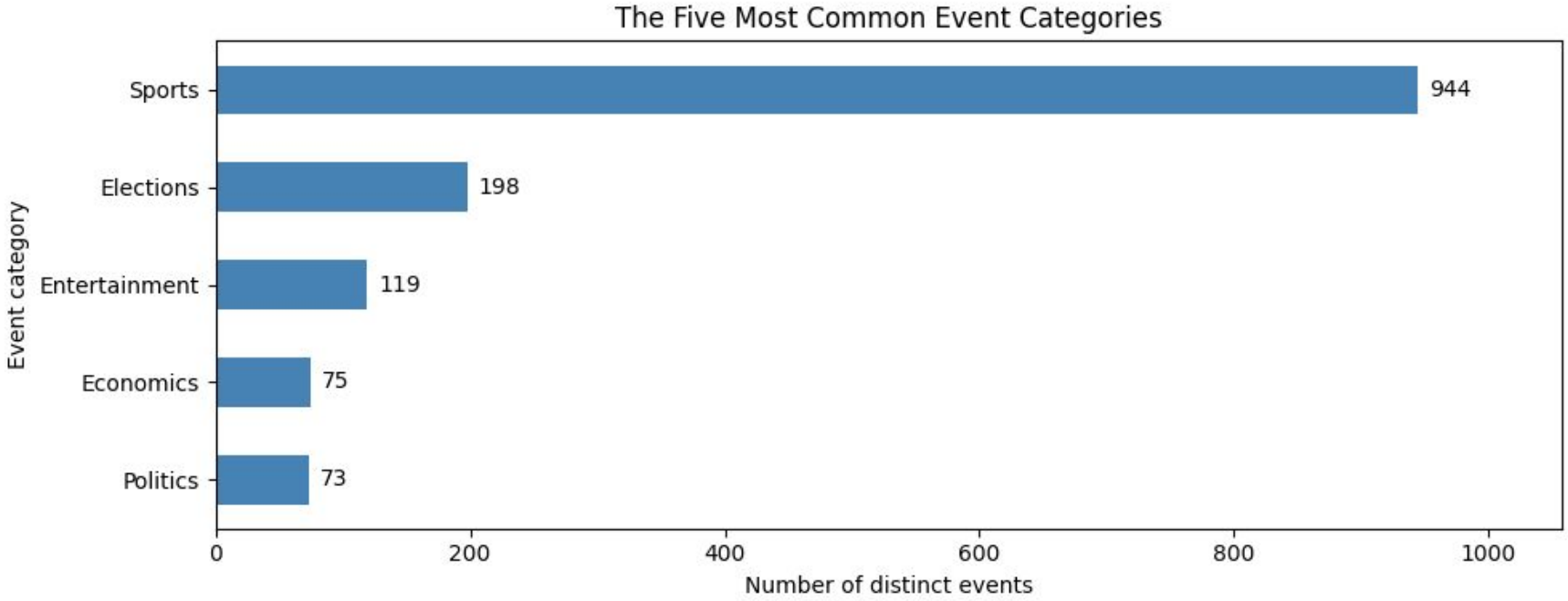
What was happening?

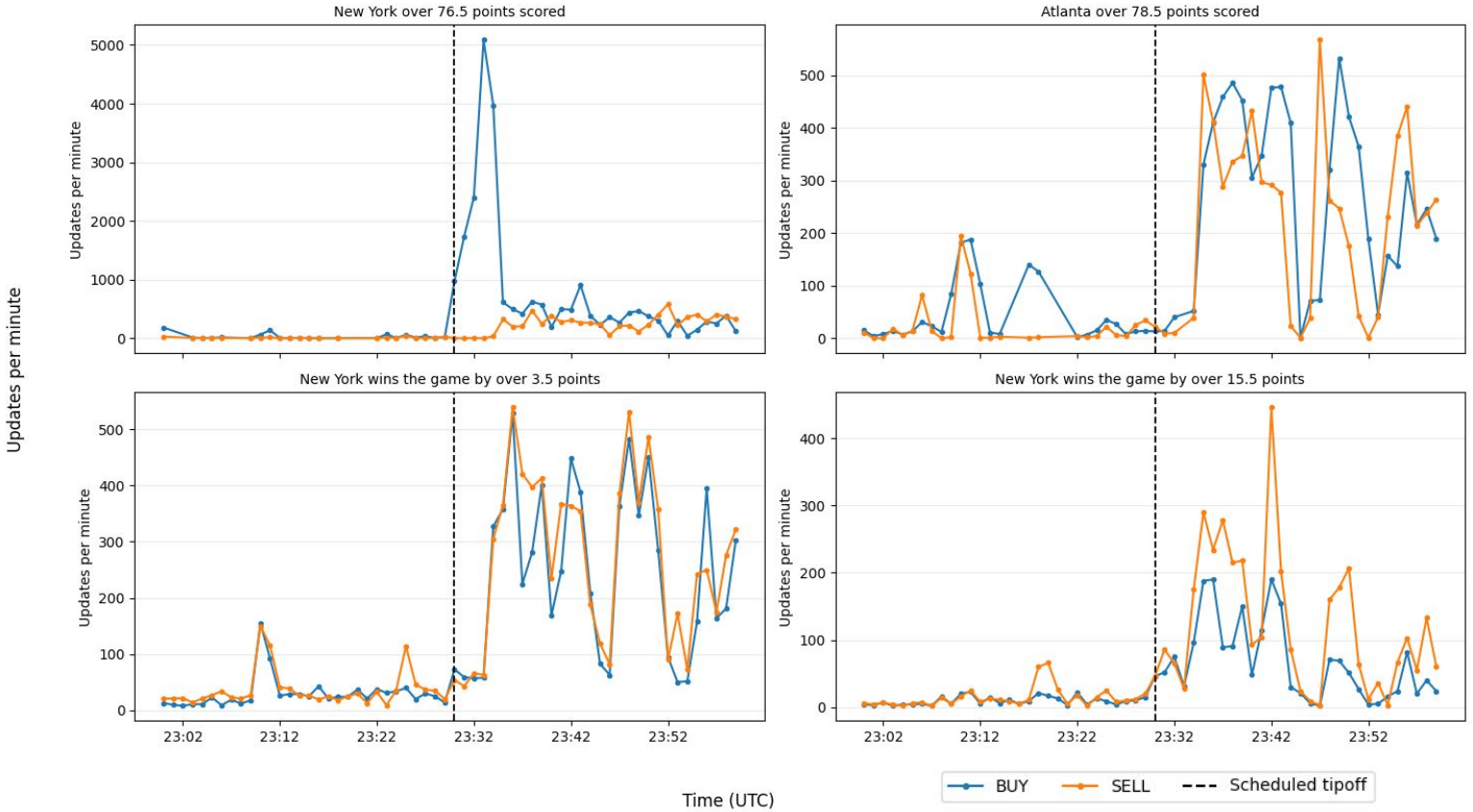
October 9th. 2026



- Order-book updates came in bursts.
- Update counts show when the book changed.

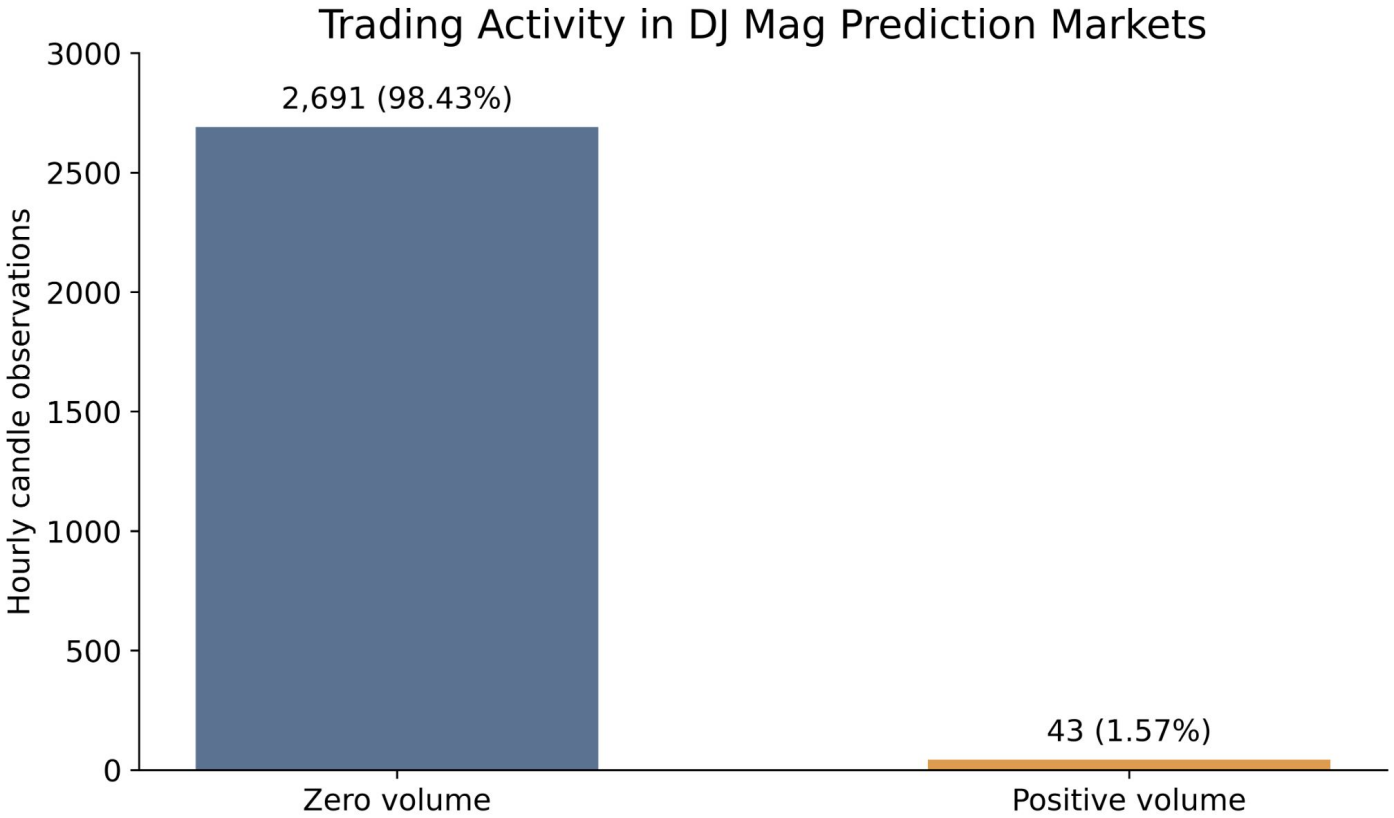
Which Event Categories Appear Most Often?



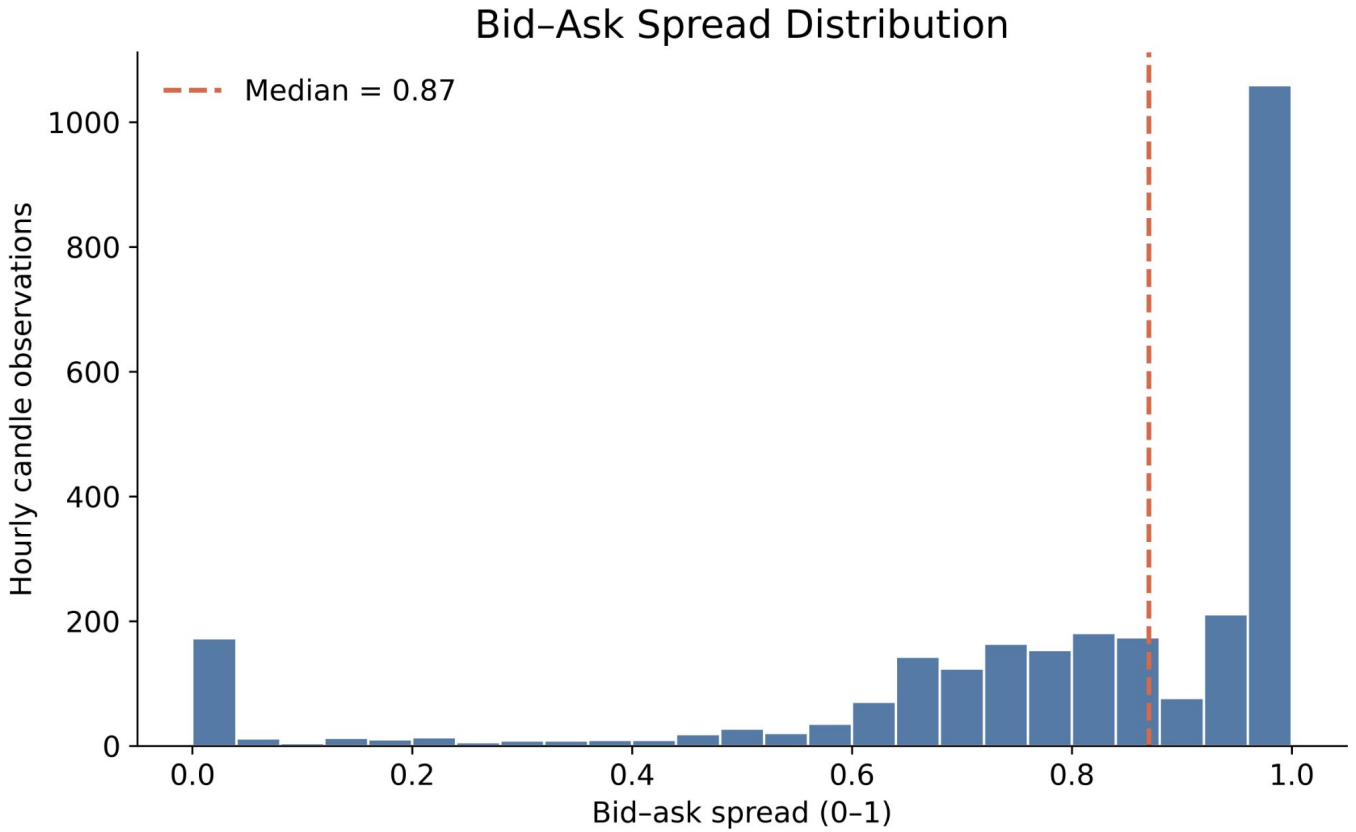


- Updates surged after tipoff, especially for New York over 76.5 points.

Niche Markets

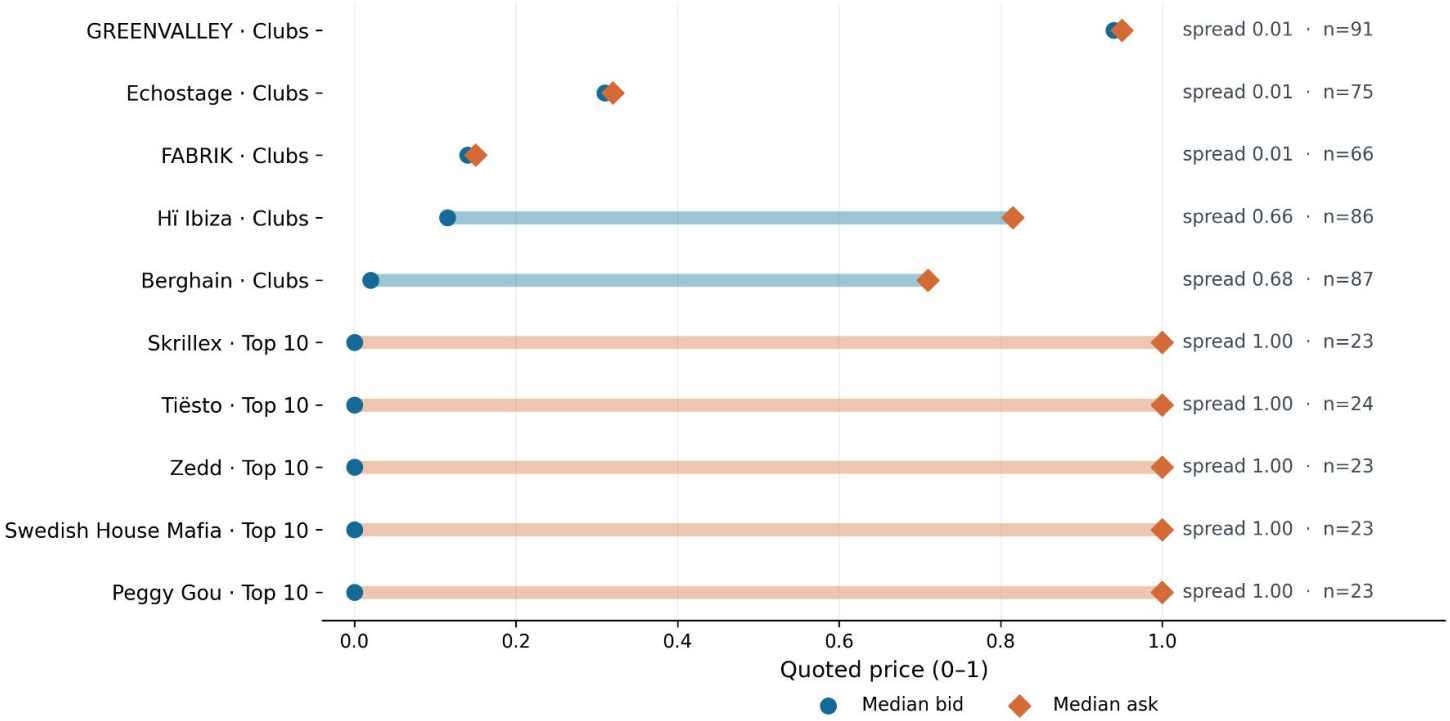


Takeaway: 98.43% of observed hourly candles recorded zero trading volume

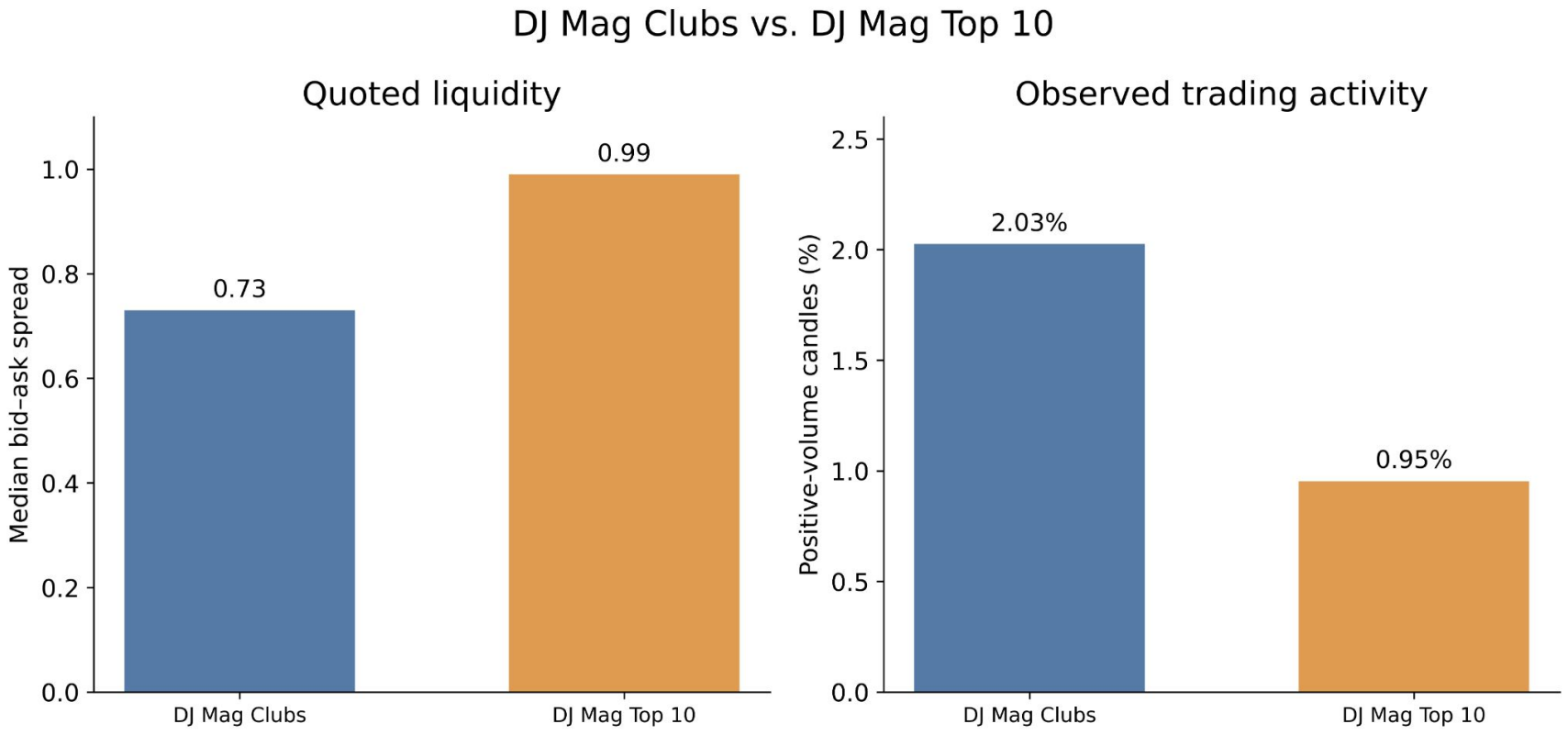


Takeaway: Median bid–ask spread of 0.87 suggests limited quoted liquidity

How Far Apart Are Buyers and Sellers?
Selected DJ Mag Prediction Markets



Dots show median bid and median ask across observed hourly candles; spread labels show median hourly spread.
Selected examples, not all 70 markets. Extreme 0/1 quotes may be missing-order placeholders.



Takeaway: DJ Mag Clubs showed tighter spreads and more frequent trading activity than Top 10